Online Supplement to "Reproducible Answer Selection over Large Language Model Ensembles, with an Exact Kemeny Reference and a Ground-Truth Audit"

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This supplement contains the extended threshold analysis and proofs, additional selectors, secondary judge and generator diagnostics, coverage, the open-ended pilot and implementation notes. Numbers prefixed with S refer to this supplement.

S1 Extended threshold analysis

This section extends Section 4 beyond $N = 3$. Unless stated otherwise, candidates are independent and, conditional on their correctness labels, comparisons are independent, same-class comparisons are fair, and top-score ties are broken uniformly. The results concern binary comparisons, not the empirical pipeline's averaged soft weights.

Lemma S1 (Monotone coupling)**.** *For every N and every ρ, the probability that the Copeland selector returns a correct candidate is nondecreasing in each q_{ij}.*

Corollary S1 (Pair-dependent judges)**.** *Let $q^*(p, N, \rho)$ be the smallest uniform q at which the selector ties voting. A judge with $q_{ij} \geq q^*$ for all mixed pairs is at least as accurate as voting; one with $q_{ij} \leq q^*$ for all mixed pairs is at most as accurate.*

Table 2 gives q^* computed exactly for $N = 3, \ldots, 7$ (the $N = 3$, $\rho = 0$ column reproduces Table 1). Two things stand out. With distinct wrong answers the computed threshold rises over $N = 3, \ldots, 7$: at $p = 0.8$ it goes from 0.804 to 0.945. Proposition S1 also gives a large-N lower bound. With correlated wrong answers ($\rho = 0.5$) voting is weaker and the threshold is lower, about 0.82 at $p = 0.8$ for $N \geq 4$. The direction of the relaxation is therefore the one a reader would expect, but its size is large enough that the realistic threshold is far from $\frac{1}{2}$. For parameters estimated from homogeneous GPQA Diamond, the model threshold falls from 0.787 to 0.714 (Table S7). The recorded soft-score mean is not an estimate of binary accuracy q. We checked the exact enumeration against an independent Monte Carlo simulation with common random numbers (Table S1; all four configurations agree within sampling error) and checked Corollary S1 on random pair-dependent judges: in 30 of 30 random $N = 5$ instances the accuracy gap lies between the gaps at $q_{\min}$ and $q_{\max}$.

Why the threshold keeps rising: a large-N lower bound With distinct wrong answers plurality's error has a closed form. It is correct whenever at least two candidates are correct, and when exactly one is correct all N answers differ and it is chosen with probability $1/N$, so

$$\mathrm{err}_V(N) = (1-p)^N + (N-1)\,p\,(1-p)^{N-1}, \tag{S1}$$

Table S1: Exact accuracy gap (Copeland minus plurality, percentage points) against a 200,000-draw Monte Carlo estimate with its standard error.

N	p	ρ	q	Exact	Monte Carlo
4	0.5	0.0	0.7	-5.40	-5.50 ± 0.11
5	0.8	0.5	0.8	-0.66	-0.68 ± 0.05
6	0.5	0.5	0.7	+2.64	+2.65 ± 0.11
6	0.8	0.0	0.9	-0.20	-0.20 ± 0.01

which decays at rate $-\ln(1-p)$ per candidate. A pairwise-Copeland selector that errs on the event "a wrong candidate beats a sizeable block of correct ones" decays more slowly unless q is large. Write $\mathrm{KL}(\eta\|p) = \eta \ln \frac{\eta}{p} + (1-\eta) \ln \frac{1-\eta}{1-p}$ and

$$\Lambda(p,q) \;=\; \inf_{1-\frac{1}{2q}<\eta\le 1} \Big[\mathrm{KL}(\eta\|p) + \eta \ln \tfrac{1}{1-q}\Big].$$

Proposition S1 (Large-N lower bound)**.** *Let wrong answers be distinct ($\rho = 0$) and the judge uniform with $\frac{1}{2} < q < 1$. If $\Lambda(p,q) < -\ln(1-p)$, there is N_0 such that for all $N \ge N_0$ the Copeland selector is strictly less accurate than plurality. Consequently* $\liminf_{N\to\infty} q^*(p,N) \ge q_\Lambda(p)$*, where $q_\Lambda(p)$ is the unique root of $\Lambda(p,q) = -\ln(1-p)$.*

Table S2 lists $q_\Lambda(p)$ next to the exact $N = 6$ thresholds: the exact values sit just below the limiting lower bound for $p \ge 0.6$ and just above it at $p = 0.5$, as a sequence rising to (at least) q_Λ would. We do not prove that the limit equals q_Λ; the proposition is a lower bound, obtained by exhibiting one specific way in which Copeland errs.

Table S2: Plurality's error decay rate $-\ln(1-p)$, the large-N lower bound $q_\Lambda(p)$ on $\liminf_N q^*(p,N)$ (Proposition S1), and the exact $q^*(p,6)$ ($\rho = 0$).

p	$-\ln(1-p)$	$q_\Lambda(p)$	$q^*(p,6)$
0.5	0.693	0.820	0.825
0.6	0.916	0.867	0.862
0.7	1.204	0.907	0.899
0.8	1.609	0.943	0.935
0.9	2.303	0.973	0.970

Structure of the wrong answers beyond a single attractor The attractor model with parameter ρ is one way to correlate wrong answers. We also drew each wrong answer uniformly from D wrong values ($D = 1$: all wrong answers coincide; $D = 3$: the three distractors of a four-option question; $D = \infty$: all distinct). Table S3 shows that the threshold is monotone in the structure: the fewer distinct wrong values, the weaker plurality and the lower the threshold, down to $q^* = \frac{1}{2}$ when all wrong answers coincide and the pool is small ($N = 4$, $p = 0.5$). For the four-option GPQA questions ($D = 3$) the threshold is 0.64–0.80 at $N = 4, 5$, illustrating sensitivity to wrong-answer structure rather than validating the attractor model for GPQA.

The cost term The thresholds concern quality-only selection. Table S4 illustrates adding independent random lengths in one simulation. A dominant cost term selects the shortest candidate; when lengths are independent of correctness and ties have probability zero, that rule has accuracy

Table S3: Exact q^* when each wrong answer is uniform over D wrong values.

	$N = 4$		$N = 5$	
D	$p = 0.5$	$p = 0.8$	$p = 0.5$	$p = 0.8$
1	0.500	0.669	0.500	0.721
2	0.594	0.754	0.592	0.770
3	0.640	0.791	0.638	0.798
8	0.710	0.847	0.723	0.854
∞	0.759	0.891	0.800	0.919

p. This limiting argument does not prove monotone accuracy degradation for every cost distribution or judge. Empirical lengths are not independent of correctness, so Section 6.4 evaluates their effect by replay.

Table S4: Simulated accuracy (%) of selection by Copeland score minus $\lambda\times$ an independent random length, $N = 3$, $p = 0.8$ (plurality voting: 92.8%; random pick: 80.0%), 100,000 draws.

λ	$q = 0.6$	$q = 0.9$
0	84.5	96.1
0.25	84.4	96.1
1	83.9	94.9
4	81.4	86.0
∞ (shortest)	79.8	79.8

Large-N consistency When both selectors approach perfect accuracy, their error rates can still differ. The following consistency result complements, rather than replaces, the threshold comparison.

Proposition S2 (Large-N regimes)**.** *Let candidates be independent with correct-answer probability $p > 0$, wrong answers taking finitely many values, and let m be the largest probability of any single wrong answer. (i) Plurality voting is correct with probability tending to 1 if $p > m$ and to 0 if $p < m$. (ii) If $q_{ij} \geq q_{\min} > \frac{1}{2}$ for all mixed pairs, the Copeland selector is correct with probability at least $1 - (1-p)^N - 2N^2 e^{-(q_{\min}-1/2)^2 N/2} \to 1$, whatever m.*

A better-than-chance judge can therefore rescue a pool when $p < m$, where plurality fails asymptotically. When $p > m$, both may be consistent; this does not determine their finite-N difference. Aggregate accuracy across prompts does not identify the regime of each prompt. The distinction parallels reward-based selection analyses (Di et al., 2026), without establishing a universal explanation for the empirical results.

S2 Proofs for the extended threshold analysis

Lemma S1. Fix a mixed pair (c, w) with c correct and w wrong and condition on all other comparison outcomes; the outcome of (c, w) is a Bernoulli(q_{cw}) variable that is the only one depending on q_{cw}, so it suffices to show that changing it from "w wins" to "c wins" cannot lower the probability that the selected candidate is correct. The change raises c's score by one and lowers w's by one. Let T be the set of top scorers before and T' after, and let the selected candidate be uniform on the top

set, so the success probability is the share of correct candidates in it. If $c \in T$, then $T' = \{c\}$, which is correct. If $c \notin T$ and $w \notin T$, then $T' = T$ or $T' = T \cup \{c\}$; adding a correct candidate cannot lower a share that is at most one. If $w \in T$ and $|T| > 1$, then T' removes w and possibly adds c, which cannot lower the share. If w is the unique top scorer, the share before is 0. In every case the share does not decrease. Averaging over the other outcomes gives the claim, and Corollary S1 follows by comparing with the uniform judges at $q_{\min}$ and $q_{\max}$ (voting does not depend on the judge).

Proposition S2. (i) The count of the correct answer is Binomial(N, p) and that of each wrong answer is Binomial$(N, \cdot)$ with success probability at most m, the largest attaining m; by Hoeffding's inequality and a union bound over the finitely many answers, the correct answer is the unique plurality with probability tending to one if $p > m$, and is beaten by the most popular wrong answer with probability tending to one if $p < m$. (ii) Condition on the set of correct candidates, of size $k_c \geq 1$ (probability $1 - (1-p)^N$), and let $k_w = N - k_c$. A correct candidate's score is a sum of $N - 1$ independent comparisons with mean at least $q_{\min} k_w + \frac{1}{2}(k_c - 1)$; a wrong candidate's has mean at most $(1 - q_{\min})k_c + \frac{1}{2}(k_w - 1)$; the difference is at least $(q_{\min} - \frac{1}{2})N$. By Hoeffding's inequality each score is within $t = (q_{\min} - \frac{1}{2})N/2$ of its mean with probability at least $1 - 2\exp(-2t^2/(N-1)) \geq 1 - 2\exp(-(q_{\min} - \frac{1}{2})^2 N/2)$. A union bound over the N candidates gives the stated probability that every correct candidate outscores every wrong one, in which case the selected candidate is correct. □

Equation (S1). With distinct wrong answers the plurality is the correct answer exactly when at least two candidates are correct (its count then exceeds every wrong count, which is one); with one correct candidate all N answers are tied and the correct one is chosen with probability $1/N$; with none it cannot be chosen. Hence $\mathrm{err}_V = (1-p)^N + Np(1-p)^{N-1}(1 - 1/N)$.

Proposition S1. Fix $q > \frac{1}{2}$ and $\eta \in (1 - \frac{1}{2q}, 1)$ and let $K = \lceil \eta N \rceil$. Consider the event E that exactly K candidates are correct and a designated wrong candidate w beats all of them (probability $\binom{N}{K} p^K (1-p)^{N-K} (1-q)^K \geq \frac{1}{N+1} e^{-N\,\mathrm{KL}(K/N \| p)} (1-q)^K$). Conditionally on E, w's score is $K + \mathrm{Bin}(N - K - 1, \frac{1}{2})$, with mean $(N + K - 1)/2$; a correct candidate's score has mean at most $q(N - K - 1) + (K - 1)/2 + 0$ (it has already lost to w), and another wrong candidate's has mean $(1-q)K + (N - K - 1)/2$. The gaps are $N(\frac{1}{2} - q(1-\eta)) + O(1)$ and qK, linear in N and positive because $\eta > 1 - \frac{1}{2q}$. By Hoeffding's inequality and a union bound over the N candidates, w is the unique top scorer with conditional probability $1 - O(Ne^{-cN})$, so the Copeland selector errs on E with probability at least $\frac{1}{2}$ for large N. Therefore $\mathrm{err}_C \geq \frac{1}{2(N+1)} e^{-N[\mathrm{KL}(\eta \| p) + \eta \ln \frac{1}{1-q} + o(1)]}$. If $\Lambda(p, q) < -\ln(1-p)$ choose η with bracket below $-\ln(1-p)$; since $\mathrm{err}_V \leq N(1-p)^{N-1}$ by Equation (S1), $\mathrm{err}_C > \mathrm{err}_V$ for all large N. For the consequence: $\Lambda(p, q)$ is continuous and strictly increasing in q (it equals an infimum of functions increasing in q over a set that shrinks with q), lies below $-\ln(1-p)$ for q slightly above $\frac{1}{2}$ (the derivative of $\mathrm{KL}(\eta \| p)$ is $-\infty$ at $\eta = 0$) and diverges as $q \uparrow 1$ (it is at least $\frac{1}{2} \ln \frac{1}{1-q}$), so $q_\Lambda(p)$ exists and is unique. For any $\epsilon > 0$ the accuracy gap at $q_\Lambda - \epsilon$ is negative for large N, and by Lemma S1 it is nonincreasing as q decreases, so it is negative for every $q \leq q_\Lambda - \epsilon$. Hence $q^*(p, N) > q_\Lambda - \epsilon$ for large N. □

S3 Additional selectors

Table S5 compares OR-Select's competitors beyond Section 6.2. Judge-free selectors (shortest and longest candidate, and minimum-Bayes-risk selection with token-set Jaccard similarity) run on every graded problem; judge-based selectors are replayed on the problems whose judge outputs were stored (pointwise confidence argmax, a vote weighted by the scorer's confidence, a knockout tournament over the pairwise matrix, and a Bradley–Terry fit). Each is paired with self-consistency

on the same problems; the parenthesis in each accuracy cell is the self-consistency accuracy on those problems.

No additional selector establishes a statistically significant improvement over plurality in these exploratory comparisons. Minimum-Bayes-risk selection is numerically ahead on some pooled comparisons; this is not evidence of equivalence or of an absence of correctness information in judge outputs. The pairwise and pointwise judge-based selectors perform worse than voting in the heterogeneous replay subset.

Table S5: Additional selectors, paired with self-consistency (SC) on the same graded problems. For each condition: problems n, accuracy % (SC accuracy % on the same problems), problems where the selector is right and SC wrong over the reverse, and the exact McNemar p. †: needs the stored judge outputs, so n is the replayable subset.

	Homogeneous				Heterogeneous			
Selector	n	Acc. (SC)	W/L	p	n	Acc. (SC)	W/L	p
Shortest candidate	532	84.6 (85.5)	10/15	0.424	402	78.4 (85.1)	11/38	<0.001
Longest candidate	532	83.3 (85.5)	11/23	0.058	402	80.1 (85.1)	5/25	<0.001
MBR (Jaccard)	532	85.9 (85.5)	9/7	0.804	402	85.3 (85.1)	9/8	1.000
Confidence argmax†	332	80.7 (80.7)	7/7	1.000	397	79.8 (84.9)	10/30	0.002
Confidence-weighted vote†	332	79.8 (80.7)	0/3	0.250	397	84.4 (84.9)	3/5	0.727
Knockout tournament†	332	79.8 (80.7)	11/14	0.690	397	79.6 (84.9)	12/33	0.002
Bradley–Terry†	332	80.1 (80.7)	10/12	0.832	397	77.8 (84.9)	8/36	<0.001

S4 Secondary empirical diagnostics

S4.1 Judge calibration and configuration history

The fixed judge's RewardBench category-mean accuracy is 60.5% (Table 3). Its micro-average is 60.9%, with 95% Wilson interval [53.2%, 68.1%] on the selected 161 pairs. Because the sample takes the first seven rows per subset rather than a probability sample, this interval does not establish representativeness for all RewardBench pairs. The pre-registration treated roughly 70–90% accuracy as support for the judge; the observed overall score is below that range.

For longer human-preferred responses, accuracy is 68.4% (54/79), with interval [57.5%, 77.6%]. For shorter preferred responses it is 53.7% (44/82), with interval [42.9%, 64.0%]. The overlapping intervals do not establish length bias. Rotating judges yields 48.5% ($n = 161$), an exploratory configuration comparison.

The pilot of 2026-08-22 used Nemotron-3-Ultra (later withdrawn) with reasoning enabled, obtaining 90.1%. On 2026-10-07 we disabled reasoning because calls often truncated (six of ten in a spot check). That configuration does not support the main audit's results.

S4.2 Generator shares and length

Table S6 describes the composite selector's choices, not the judge in isolation. Cohere supplies 75.6% of winners, but its selected candidates are correct on 79.6%, versus 83.9% across all its candidates. Generator, length, confidence and quality effects are not separated by this table. The winner is the longest candidate on 1% of heterogeneous pools and 6% of homogeneous pools; its

length is on average 0.38 and 0.70 times the corresponding pool mean. Pool composition changes are reported in Table S8.

Table S6: Generator composition and OR-Select picks over 402 heterogeneous problems. Accuracy and pick share are percentages.

Model	Candidates	Cand. acc.	OR-Select picks	Pick acc.
Nemotron-3.5 (judge model)	804	82.5	21 (6.1%)	90.5
North-Mini-Code	802	83.9	260 (75.6%)	79.6
Laguna-S 2.1	548	73.4	53 (15.4%)	67.9
Xing4.0-29B	256	56.2	9 (2.6%)	44.4
LFM-2.5 2.6B	2	50.0	1 (0.3%)	100.0

S4.3 Soft scores and model sensitivity

Table S7 retains the descriptive score summary and model sensitivity calculations. Here $\hat{p}$ uses all valid stored candidate pools; $\bar{W}_c$ is the mean soft weight assigned to the correct candidate over mixed pairs in rows retaining judge outputs. Each unordered mixed pair contributes once, in its correct-to-wrong direction. These rows are not a random subset; there are only 28 mixed pairs for homogeneous MMLU-Pro and 72 for homogeneous GPQA. Pairs within a problem are clustered.

The wrong-answer attractor parameter is estimated as the square root of wrong-pair agreement; 400 problem-level bootstrap resamples give an interval for this parameter alone. The table plugs $\hat{p}$ and $\hat{\rho}$ into the binary model. The resulting range does not incorporate uncertainty in p, judge accuracy, score calibration or model mismatch. It is not a confidence interval for a deployment decision.

Crucially, $\bar{W}_c$ is not the Bernoulli accuracy q of Theorem 4.1. For example, weights 0.6 on every correct-to-wrong pair and 0.5 within each correctness class make quality-only selection choose a correct candidate whenever one exists, despite a correct-directed mean of 0.6. A binary comparison with success probability 0.6 is a different experiment. Neither the soft mean nor its distance from the tabulated thresholds establishes that the actual selector, or every selector using these outputs, must lose.

Table S7: Descriptive soft scores and binary-model sensitivity, not an empirical threshold test. $\bar{W}_c$: correct-directed mixed-pair weight; $\hat{\rho}$: square root of wrong-pair agreement, with 95% bootstrap interval. q_0^* and $q_{\hat{\rho}}^*$ are binary-model thresholds; the bracketed range propagates only the ρ interval. ‡: fewer than 30 wrong pairs, so the full $\rho \in [0, 1]$ range is used. H denotes heterogeneous generation.

Track	N	$\hat{p}$	Wrong pairs	$\hat{\rho}$ [95% CI]	Mixed pairs	$\bar{W}_c$	q_0^*	$q_{\hat{\rho}}^*$
GPQA Diamond	3	0.771	94	0.88 [0.82, 0.93]	72	0.536	0.787	0.714 [0.706, 0.724]
MMLU-Pro	3	0.850	31	0.84 [0.66, 0.96]	28	0.443	0.836	0.786 [0.772, 0.805]
GSM8K-H	6	0.975	3	n/a	24	0.660	0.995	[0.904, 0.995]‡
MATH L5-H	6	0.818	219	0.66 [0.50, 0.80]	292	0.536	0.941	0.789 [0.755, 0.828]
AIME 2026-H	6	0.767	32	0.31 [0.00, 0.47]	146	0.560	0.923	0.857 [0.809, 0.923]
GPQA Diamond-H	6	0.718	530	0.82 [0.77, 0.87]	615	0.489	0.905	0.682 [0.667, 0.697]

S5 Coverage and higher-cap second pass: tables

Table S8: Coverage by track. Truncated: no valid answer after resampling because generation hit the output cap. Other: failed for another reason. Untried: not attempted. With Laguna: graded heterogeneous problems that still contain Laguna-S candidates.

Track	Total	Graded	Truncated	Other	Untried	With Laguna
B — GSM8K, homog.	40	40	0	0	0	–
B′ — GSM8K, heterog.	40	40	0	0	0	40
C — MATH L5, homog.	134	134	0	0	0	–
C′ — MATH L5, heterog.	134	134	0	0	0	131
D — AIME 2026, homog.	30	30	0	0	0	–
D′ — AIME 2026, heterog.	30	30	0	0	0	23
E — GPQA Diamond, homog.	198	198	0	0	0	–
E′ — GPQA Diamond, heterog.	198	198	0	0	0	80
F — AIME 2025, homog.	30	30	0	0	0	–
G — MMLU-Pro, homog.	100	100	0	0	0	–

Table S9: Effect of the flagged second pass. Main protocol: problems that completed under the 32,768-token cap. Rescued: problems completed only under the higher cap. Final: all graded problems.

Track	Main n	Main OR	Main SC	Rescued	Final OR	Final SC
C — MATH L5, homog.	132	90.9	92.4	2	89.6	91.0
C′ — MATH L5, heterog.	131	79.4	90.8	3	79.1	90.3
D — AIME 2026, homog.	24	95.8	91.7	6	90.0	86.7
D′ — AIME 2026, heterog.	23	78.3	100.0	7	63.3	93.3
E — GPQA Diamond, homog.	175	82.3	82.9	23	78.8	78.8
E′ — GPQA Diamond, heterog.	163	85.3	82.8	35	79.3	77.3
F — AIME 2025, homog.	24	87.5	95.8	6	76.7	83.3
G — MMLU-Pro, homog.	94	88.3	88.3	6	86.0	86.0

S6 Pilot illustration: open-ended queries

Three open-ended queries ($N = 5$ each, no ground truth) were run in an earlier pilot to check that the pipeline behaves sensibly where self-consistency has nothing to vote on. Across the 15 candidates, none was byte-identical to another, so self-consistency resorted to an arbitrary tie-break on every query, and OR-Select differed from both baselines on all three (score margins 0.088, 0.446 and a near-tie of 0.003). With three queries and no ground truth this is an illustration, not evidence about quality. Two pilot ablations on the same candidate sets (setting $w_4 = 0$; forward-only judging) changed 0 and 2 of the 3 winners respectively: the self-consistency term was inert because no candidate set clustered, and bidirectional judging mattered in two of three cases, consistent with position bias being a first-order judge failure mode (Zheng et al., 2023; Tang et al., 2024; Wang et al., 2024). Three instances cannot characterize when the directions disagree.

S7 A judge-output stability observation

In an earlier pilot run on a different provider and model, three candidates had byte-identical text. Forward/backward averaging gives 0.5 only if the two raw scores coincide; identical prompts alone do not ensure this. The observed quality scores were 0.375, 0.50 and 0.625, showing variation across the recorded judge calls, consistent with reported run-to-run instability (Atil et al., 2025). This observation alone does not isolate seed handling as the cause. The exact-match short-circuit (Section 3.1) removes this failure mode for identical text but not for different texts that receive unstable scores, which remains open.

S8 Retries and extraction fallbacks

Each validity retry uses a freshly varied seed to avoid requesting an identical repeat; this does not establish statistical independence. Extraction accepts more than one marker: GPQA also recognizes a prose "the answer is X" pattern alongside `ANSWER: X` and `\boxed{X}`, because its answer space is four letters; MATH's open-ended `\boxed{}` extraction has no prose fallback, since a wrong guess there would convert an honest "ungraded" problem into a confidently incorrect one. This asymmetry is a design choice.

S9 Worked example and positional-bias figure

Table S10 and Figure S1 decompose Equation (1) for one real open-ended run ($N = 5$, no exact-text duplicates). quality_i is recovered by inverting the saved score_i against the known confidence, cost and consistency, and the recovered argmax is checked by script against the saved winner.

Table S10: Component scores for one real open-ended run (winner marked *). quality_i is recovered by inversion, not stored directly.

Candidate	quality_i	confidence_i	consistency_i	cost_i	score_i
c0	0.606	1.00	0.20	38	0.486
c1	0.387	1.00	0.20	32	0.327
c2*	0.644	1.00	0.20	33	0.574
c3	0.575	1.00	0.20	38	0.455
c4	0.288	1.00	0.20	33	0.218

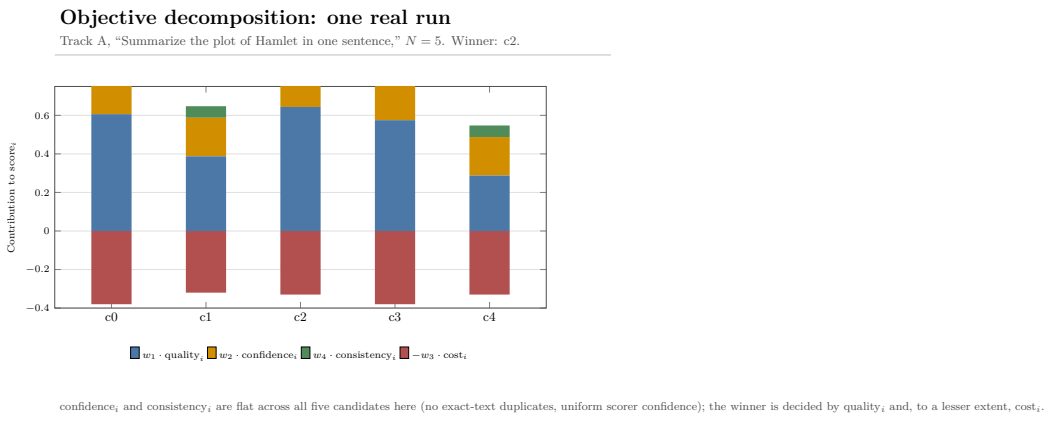


Figure S1: The same run as Table S10, as stacked weighted contributions to score$_i$. Confidence and consistency are constant across candidates here (no duplicates; uniform scorer confidence), so the ranking is decided by quality and cost. These candidates are one-sentence summaries; on multi-hundred-word reasoning outputs the cost term is far larger relative to quality (Section 6.4).

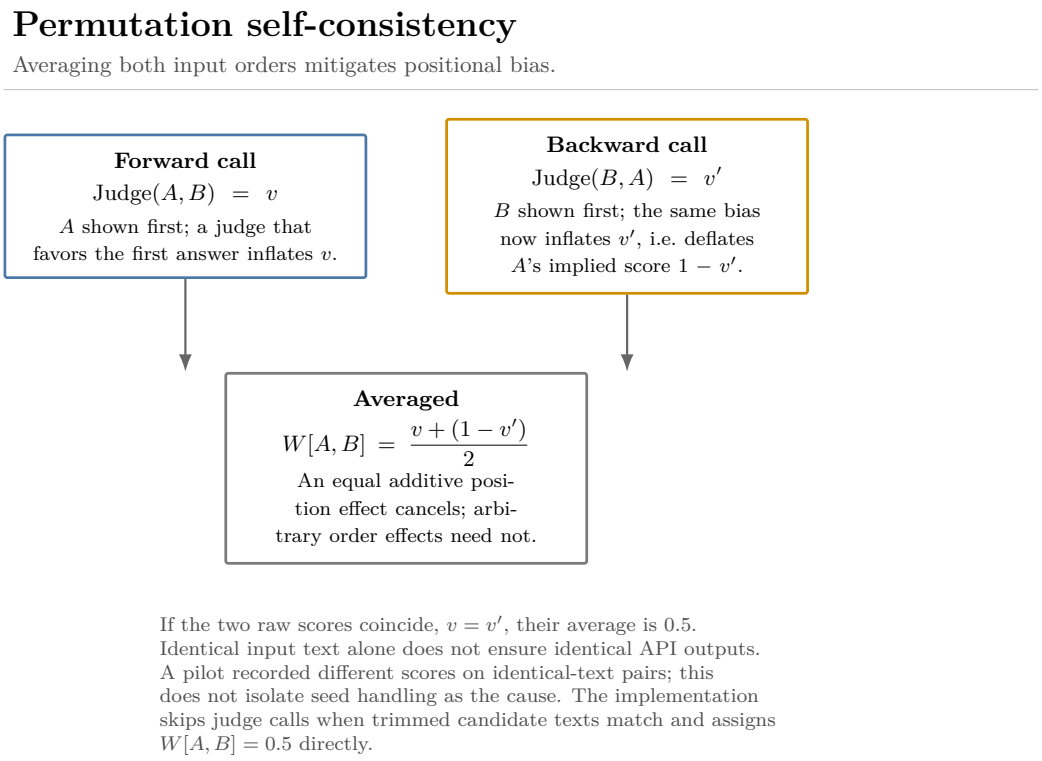


Figure S2: Forward/backward averaging cancels equal additive positional effects, not arbitrary order dependence. For identical trimmed candidate texts, the implementation skips both calls and directly assigns $W = 0.5$.

References

Atil, B., Aykent, S., Chittams, A., Fu, L., Passonneau, R. J., Radcliffe, E., Rajagopal, G. R., Sloan, A., Tudrej, T., Ture, F., Wu, Z., Xu, L., & Baldwin, B. (2025). Non-determinism of "deterministic" LLM system settings in hosted environments. *Proceedings of the 5th Workshop on Evaluation and Comparison of NLP Systems*, 135–148. https://doi.org/10.18653/v1/2025.eval4nlp-1.12

Di, Q., Ji, K., Li, X., Zhao, H., & Gu, Q. (2026). Best-of-majority: Minimax-optimal strategy for pass@k inference scaling. *International Conference on Learning Representations (ICLR)*. https://proceedings.iclr.cc/paper_files/paper/2026/hash/94cbc452f8fe10cc6350e3c06fe75171-Abstract-Conference.html

Tang, R., Zhang, X., Ma, X., Lin, J., & Ture, F. (2024). Found in the middle: Permutation self-consistency improves listwise ranking in large language models. *Findings of the Association for Computational Linguistics: NAACL 2024*.

Wang, P., Li, L., Chen, L., Cai, Z., Zhu, D., Lin, B., Cao, Y., Kong, L., Liu, Q., Liu, T., & Sui, Z. (2024). Large language models are not fair evaluators. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 9440–9450. https://doi.org/10.18653/v1/2024.acl-long.511

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems (NeurIPS)*.